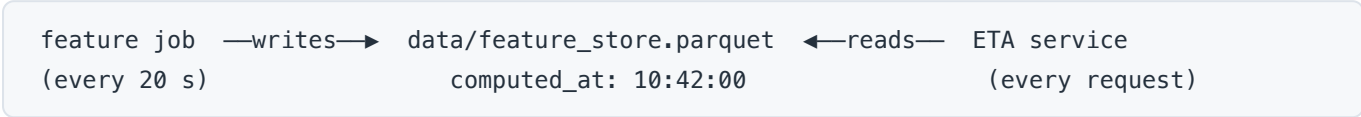


Slot 6 · Freeze your own features

File: `freshness_check.py` · **Done when:** `uv run pytest tests/test_slot06.py` shows 2 passed (service must be running) · **Time:** 15 min

Today's idea

Two programs share one file, and nothing else:



If the feature job dies, the service does not know. It keeps answering with **old** data. Old data is not "wrong" — so every normal check still passes. Today you write the check that catches it.

Step 1 · Write the check

Open `freshness_check.py`. Write `check_freshness(max_age):`

- 1. Ask the service: `info = httpx.get(f"{URL}/store-info", timeout=5).json()`
- 2. Work out the age: now (in UTC) minus `datetime.fromisoformat(info["computed_at"])`, in seconds
- 3. If the age is more than `max_age`, raise `FeatureStaleness(...)` with a clear message
- 4. Otherwise return `{"age_seconds": ..., "rows": info["rows"]}`

Try it (the service must be running — `make serve`):

```
uv run python freshness_check.py
```

```
{'age_seconds': 44, 'rows': 900}
```

Tip Use `datetime.now(timezone.utc)`. Plain `datetime.now()` has no timezone and Python will refuse to subtract.

Step 2 · Three terminals

Open three Terminal windows, all inside `eta-mlops`. Put them side by side.

terminal	command	what it does
T1	<code>uv run python -m eta.feature_job --loop 20</code>	refreshes the store every 20 s
T2	<code>make serve</code>	the ETA service (skip if already running)
T3	<code>uv run python freshness_check.py --watch --max-age 45</code>	predicts + checks every 10 s

In T3 you should see `HTTP 200`, the same ETA, and `ok` with a small age.

Step 3 · Kill the feature job

Click on T1 and press **Ctrl + C**. Now just watch T3 for about one minute.

```
t http      eta freshness check
40  200      45.1 ok      {'age_seconds': 42, 'rows': 900}
50  200      45.1 FIRED  feature store is 52 s old (limit 45 s)
60  200      45.1 FIRED  feature store is 62 s old (limit 45 s)
```

Think The ETA did not change. The status is still 200. Without your check, who would have noticed? When?

Press **Ctrl + C** in T3 to stop watching.

Step 4 · A faster way to make it stale

```
uv run python -m eta.corrupt freeze --hours 3
curl -s -X POST localhost:8000/reload-store
uv run python freshness_check.py
```

You should see `FeatureStaleness: feature store is 10800 s old (limit 120 s)`.

Now put it back and check again:

```
uv run python -m eta.corrupt restore
curl -s -X POST localhost:8000/reload-store
uv run python freshness_check.py
```

Step 5 · Check you are done

```
uv run pytest tests/test_slot06.py
```

Think Who in a company should decide the number 120 seconds — the engineer or the product manager?

If something goes wrong

what you see	what to do
tests show <code>skipped</code>	The service is not running. <code>make serve</code>
check never fires	The service is caching the old store. Run the <code>reload-store</code> curl.
check fires immediately	<code>max_age</code> is too small for a 20 s loop.
<code>can't subtract offset-naive and offset-aware</code>	Use <code>datetime.now(timezone.utc)</code> .
T1 is not updating	T1 must be inside the <code>eta-mlops</code> folder.